

Supplementary material S13

Snow patch identification based on Sentinel-2 data.

To identify the best performing method for snow patch identification based on Sentinel-2 data, the same points used for the plausibility assessment of the BST method (Supplementary material S5) were manually assigned as snow / no snow, based on optical interpretation of available Sentinel-2 scenes during the 2021/22 to 2022/23 austral summer seasons (2). Afterwards snow patches were identified using four different classification methods:

- Classification based on the NDSI using 0.4 as threshold (NDSI)
- NDSI scene based threshold identification using the adjusted BST approach and 0.4 as a fall back in case of unimodal distributions of the NDSI (NDSI_BST)
- 10 – The BST (Blue_BST)
- The snow classification provided directly in the Sentinel-2 L2A data

Afterwards the F1 score was calculated for all methods using the Afterwards the F-score was calculated for both situations with the 'F_meas' function provided in the 'caret' package (Tab. 1).

15 Finally, the best performing methods was identified by comparing F1 scores across all dates and using the method which outperformed the other in the majority of dates (Tab. 2). Consequently, the NDSI_BST method was used for snow patch identification based on Sentinel-2 scenes.

Table 1. Overview of F1 score results across dates and methods.

Date	F1_NDSI	F1_NDSI_BST	F1_BLUE_BST	F1_S2_snow
2021-12-28	0.68	0.95	0.76	0.51
2022-01-07	0.77	0.89	0.86	0.81
2022-01-08	0.73	0.93	0.84	0.62
2022-01-16	0.74	0.93	0.79	0.63
2022-01-27	0.73	0.93	0.92	0.76
2022-01-28	0.90	0.93	0.93	0.86
2022-11-22	0.98	0.89	0.78	0.52
2022-11-24	0.62	0.82	0.82	0.48

Table 2. Summary of best performing methods.

Method	Wins	%
NDSI_BST_ref	6	75.00
BLUE_BST_ref	2	25.00
NDSI	1	12.50
S2_Snow	0	0.00